

FITTING GAPS AND EFFICIENT OPTIMIZATION IN MULTI-TEACHER ON-POLICY DISTILLATION

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ABSTRACT

Multi-teacher on-policy distillation (MOPD) trains one student from several specialist teachers. The student shares its parameters, optimizer state, and training budget among all tasks. Its remaining teacher loss can come from local changes that are unreachable even task by task, additional conflict from requiring one shared update, and finite training that stops before the student reaches the best available local fit.

We introduce a local diagnostic that separates these three sources into a solo fitting gap, a shared-update conflict gap, and an optimization gap. The same model predicts how a small change in one task weight redistributes teacher loss among the tasks. The predicted loss response gives a direct test of teacher trade-offs in short training branches. The analysis also separates task weights, which define the intended teacher-loss mixture, from task sampling probabilities, which allocate gradient evaluations.

The separation between weights and sampling leads to Gradient-Preserving Adaptive Sampling (GPAS). GPAS samples tasks in proportion to their target weights and optimizer-scaled gradient signals, then applies an importance correction. The correction keeps the expected gradient equal to the gradient of the chosen MOPD objective. Our experiments test the local diagnostic, the effect of AdamW state on task update scale, and the training efficiency and mixture drift of GPAS under a matched token and teacher-compute budget.

1 INTRODUCTION

Large language models often use separate post-training runs for mathematics, code, instruction following, science, and other capabilities. Multi-teacher on-policy distillation (MOPD) combines these specialists in one student. Each student prompt is sent to a task teacher, and the student learns from dense teacher feedback on its own responses (Ma et al., 2026a).

Correct routing identifies the source of supervision. Shared student parameters still connect the tasks. An update from one teacher can change the student on other tasks, and a finite training budget can give some tasks more effective optimization than others. Recent MOPD studies report both mixture-dependent capability changes (Li et al., 2026b) and large differences in token exposure and learning progress (Gao et al., 2026).

A high teacher loss has two broad interpretations: local fitting error and unfinished optimization. The fitting error itself can arise because a teacher’s requested change is outside the student’s local reachable subspace, or because several individually reachable changes cannot be realized by one shared update. In the latter case, task weights choose how the student distributes the remaining conflict. Alternatively, the requested changes may admit a good shared fit while finite optimization stops short of it. More training or a better allocation of updates can reduce this last source. A useful MOPD diagnosis should distinguish these cases before changing the training recipe.

We study the semi-gradient update used by reverse-KL MOPD. At a training checkpoint, we freeze a batch of student prefixes and linearize the change in student outputs. The resulting local problem asks whether one parameter update can fit the teacher–student differences for all tasks. Comparing the joint fit with separate per-task fits divides the residual into a solo fitting gap and a shared-update conflict gap. The distance between the realized update and the best local fit defines the optimization

gap. We also derive the response to a small task-weight change and test the predicted per-task loss changes with short branches from the same checkpoint and optimizer state.

The diagnostic leads to a simple training rule. Task weights specify the teacher-loss mixture that the student should fit. Task sampling probabilities specify where the training system spends gradient evaluations. Importance correction allows an adaptive sampler to change the second quantity while keeping the expected target gradient fixed. We use this fact to derive Gradient-Preserving Adaptive Sampling (GPAS). For a fixed-token task batch, GPAS assigns more samples to tasks with larger optimizer-scaled gradient signals. The sampled gradient is rescaled before the AdamW state update.

Our study makes three contributions:

- We separate solo local reachability, shared-update conflict, and finite-step optimization error and turn the decomposition into a checkpoint-level training diagnostic.
- We predict how small task-weight changes redistribute per-task teacher loss and verify the prediction with matched short training branches.
- We derive and evaluate GPAS as a low-overhead allocation rule for a fixed MOPD objective. The evaluation measures gradient-estimator error, teacher-loss efficiency, and drift in the final teacher trade-off.

Section 3 reviews the standard MOPD objective and importance-corrected task sampling. Section 4 then states our local fitting, weight-response, and optimizer-aware allocation theory; each of these predictions is tied to a corresponding experiment in Section 5.

2 RELATED WORK

Multi-teacher on-policy distillation. MOPD trains a shared student from routed domain teachers on student-generated responses (Ma et al., 2026a). Related systems use task-specific soft prompts with a shared teacher backbone (Ma et al., 2026b) and language-specific teachers for speech recognition (Xie et al., 2026). Open-MOPD identifies unequal response lengths, teacher–student gaps, and stale rewards as sources of training imbalance (Gao et al., 2026). Its gap-following weights adapt the effective teacher mixture during training. Our analysis keeps the intended mixture explicit and studies adaptive sampling with an importance correction.

Teacher influence and multi-task updates. Teacher effects can extend beyond the routed prompt domain. Changing the teacher data mixture can move several student capabilities toward the teacher with greater exposure (Li et al., 2026b). Our local weight response predicts this movement at the teacher-loss level. Work on multi-task SFT and RL studies parameter-update magnitude and gradient similarity (Zhu et al., 2026). We measure a related quantity with a closer connection to MOPD training: the change in each task’s teacher loss after a weight perturbation. We also measure the task-dependent parameter step created by the shared AdamW state.

Adaptive task allocation. Multi-task methods often adjust loss weights to balance gradient magnitudes (Chen et al., 2018). Recent multi-task RL methods also adapt task sampling to improve weak tasks (Corrado et al., 2026). Importance sampling changes the sampling distribution and rescales sampled gradients to estimate a fixed target gradient with lower variance (Alain et al., 2015; Katharopoulos & Fleuret, 2018). GPAS applies this principle to fixed-token, task-level batches in MOPD and measures the effect of AdamW’s preconditioner on the allocation score.

3 PRELIMINARIES: MULTI-TEACHER ON-POLICY DISTILLATION

This section fixes the standard MOPD objective and the distinction between task weights and task-sampling probabilities. These definitions are problem setup rather than contributions of this work. Our checkpoint-level theory begins in Section 4.

3.1 MOPD OBJECTIVE

Consider M tasks. Task i has a fixed teacher τ_i . At a student checkpoint θ_0 , let $\mu_i^{\theta_0}$ be the distribution of student-generated prefixes for that task. Following the usual semi-gradient MOPD update, the prefixes are treated as fixed while differentiating the reverse-KL loss (Ma et al., 2026a):

$$\mathcal{L}_w(\theta; \theta_0) = \sum_{i=1}^M w_i \mathbb{E}_{c \sim \mu_i^{\theta_0}} [D_{\text{KL}}(\pi_\theta(\cdot | c) \parallel \tau_i(\cdot | c))], \quad (1)$$

where $w_i > 0$ and $\sum_i w_i = 1$. Each task loss is averaged over its valid response tokens before applying w_i . Thus, w specifies the intended teacher-loss mixture independently of response length and dataset size. Our experiments use the sampled-token reverse-KL estimator implemented by the training code.

3.2 TASK WEIGHTS VERSUS TASK SAMPLING

Let G_i be a stochastic gradient of task i 's mean-token loss and let $g_i = \mathbb{E}[G_i]$. The target gradient of Equation 1 is

$$g_w = \sum_{i=1}^M w_i g_i. \quad (2)$$

Suppose the training system samples task I with probability a_I . The standard importance-corrected estimator (Alain et al., 2015; Katharopoulos & Fleuret, 2018)

$$\hat{g} = \frac{w_I}{a_I} G_I, \quad I \sim a, \quad (3)$$

satisfies

$$\mathbb{E}[\hat{g}] = g_w. \quad (4)$$

The weights w therefore define the optimization objective, whereas the probabilities a allocate stochastic gradient evaluations. Without the factor w_I/a_I , the sampler instead follows $\sum_i a_i g_i$ and uses its sampling distribution as the effective teacher mixture.

Our sampling unit is one task-homogeneous microbatch with a fixed number of valid response tokens. Each task loss is normalized before applying w_I/a_I . Fixed-token task batches align the sampling unit with the task-level objective and prevent response length from implicitly setting the task weight.

4 OUR THEORY: LOCAL FITTING AND EFFICIENT ALLOCATION

We now introduce the checkpoint-level analysis evaluated in Section 5. The analysis has two parts. First, a local sampled-token fit separates solo reachability, shared-update conflict, and finite-update optimization error, then predicts the response to task weights. Second, a fixed-objective sampling analysis yields the optimizer-aware allocation used by GPAS.

4.1 A LOCAL QUADRATIC MODEL OF MOPD

Fix a checkpoint θ_0 . Let $\nu_i^{\theta_0}$ denote the distribution of valid response-token pairs $z = (c, y)$ obtained from task- i student rollouts at that checkpoint. For one such pair, define the sampled-token teacher signal and its score gradient,

$$d_i(z) = \log \tau_i(y | c) - \log \pi_{\theta_0}(y | c), \quad j_i(z) = \nabla_\theta \log \pi_{\theta_0}(y | c). \quad (5)$$

For compactness, let $J_i \Delta$ denote the scalar function $z \mapsto j_i(z)^\top \Delta$, and define

$$\|h\|_i^2 = \mathbb{E}_{z \sim \nu_i^{\theta_0}} [h(z)^2]. \quad (6)$$

In experiments this expectation is a mean over cached tokens. J_i is only a matrix-free linear map evaluated with Jacobian–vector and vector–Jacobian products; no token-by-parameter Jacobian is stored.

Linearizing the sampled student log probability on the fixed cache gives $\log \pi_{\theta_0 + \Delta}(y \mid c) = \log \pi_{\theta_0}(y \mid c) + j_i(z)^\top \Delta + O(\|\Delta\|^2)$. We therefore use the empirical Gauss–Newton surrogate

$$L_w(\Delta) = \frac{1}{2} \sum_{i=1}^M w_i \|d_i - J_i \Delta\|_i^2. \quad (7)$$

This model fits the same sampled-token log-ratio signal used by training. It is a checkpoint-level empirical diagnostic, not an exact representation of the full-vocabulary KL or a global model of the training trajectory. Appendix A.1 gives its construction and its local connection to reverse KL.

4.2 OUR FITTING-GAP DECOMPOSITION

Let Δ_w^* minimize the unconstrained local surrogate in Equation 7, and let $L_w^* = L_w(\Delta_w^*)$. The weighted least-squares geometry gives the following decomposition.

Proposition 1 (Local fitting and optimization error). *For every local update Δ ,*

$$L_w(\Delta) = L_w^* + \frac{1}{2} \sum_{i=1}^M w_i \|J_i(\Delta - \Delta_w^*)\|_i^2. \quad (8)$$

The first term is the residual of the best shared update within this local model. The second is the additional local error from not reaching that update.

Appendix A.2 proves the proposition. The residual L_w^* alone does not identify cross-task conflict: it can remain positive even for one task whose requested change is outside the student’s local tangent space. To isolate the cost of sharing an update, define

$$\ell_i(\Delta) := \frac{1}{2} \|d_i - J_i \Delta\|_i^2, \quad (9)$$

$$L_{i,\text{solo}}^*(R) := \min_{\|\Delta_i\|_2 \leq R} \ell_i(\Delta_i), \quad L_{w,\text{solo}}^*(R) := \sum_i w_i L_{i,\text{solo}}^*(R), \quad (10)$$

$$L_w^*(R) := \min_{\|\Delta\|_2 \leq R} \sum_i w_i \ell_i(\Delta). \quad (11)$$

The joint optimum obeys $L_w^*(R) \geq L_{w,\text{solo}}^*(R)$ because separate fits can choose a different update for each task. For $L_w(0) > 0$ and $\|\Delta_{\text{train}}\|_2 \leq R$, our diagnostic is

$$\text{SoloGap}(R) := \frac{L_{w,\text{solo}}^*(R)}{L_w(0)}, \quad (12)$$

$$\text{ConflictGap}(R) := \frac{L_w^*(R) - L_{w,\text{solo}}^*(R)}{L_w(0)}, \quad (13)$$

$$\text{FitGap}(R) := \frac{L_w^*(R)}{L_w(0)} = \text{SoloGap}(R) + \text{ConflictGap}(R), \quad (14)$$

$$\text{OptGap}(R) := \frac{L_w(\Delta_{\text{train}}) - L_w^*(R)}{L_w(0)}. \quad (15)$$

These quantities satisfy $\text{SoloGap}(R) + \text{ConflictGap}(R) + \text{OptGap}(R) = L_w(\Delta_{\text{train}})/L_w(0)$ and are nonnegative. The solo gap measures residual fit error when each task gets its own update in the tested local model and radius. It is not a claim about global student capacity. The conflict gap is the additional local residual caused by requiring one shared update. These quantities depend on the cache, parameterization, and radius; we therefore report held-out estimates, solver diagnostics, and constraint status. Exact nonnegativity holds for the objective on which the solutions are defined; cross-fitted estimates can deviate slightly because of finite cache error.

4.3 OUR PREDICTION FOR TASK-WEIGHT INTERVENTIONS

The residual in the best local fit also determines how a change in the target mixture moves each task. Let $\bar{\Delta}_w$ denote the joint solution used for the diagnostic: Δ_w^* when the solution is interior, and

arg min $_{\|\Delta\|_2 \leq R} L_w(\Delta)$ when the trust-region boundary is active. Define

$$H_w = \sum_i w_i \mathbb{E}_i[j_i(z)j_i(z)^\top], \quad r_i = \mathbb{E}_i[j_i(z)(d_i(z) - j_i(z)^\top \bar{\Delta}_w)], \quad (16)$$

where $\mathbb{E}_i$ is over $z \sim \nu_i^{\theta_0}$. Here H_w denotes a positive-semidefinite linear operator, not a matrix that is formed or inverted. Parameterize the weights as $w = \text{softmax}(\eta)$ and let $u_i = \partial \bar{\Delta}_w / \partial \eta_i$. For an interior solution, the response is obtained from the matrix-free linear solve

$$H_w u_i = w_i r_i. \quad (17)$$

For a damped interior solution, the implementation instead solves $(H_w + \xi I)u_i = w_i(r_i - \xi \bar{\Delta}_w)$. If the radius is active, we differentiate the trust-region optimality conditions. All systems are solved with matrix-free conjugate gradients; Appendix A.2 gives the details. For the local task loss $\ell_j(\Delta) = \frac{1}{2} \|d_j - J_j \Delta\|_j^2$, a small intervention $\delta \eta_i$ therefore gives

$$\delta \ell_j \approx -r_j^\top u_i \delta \eta_i. \quad (18)$$

The cross-task inner products in Equation 18 predict which other sampled-token teacher losses improve or deteriorate. Section 5.2 uses the interior or active-constraint solve according to the reported constraint status and tests the full response matrix with matched short branches.

4.4 OUR OPTIMIZER-AWARE ALLOCATION RULE

The preceding analysis concerns the target mixture w . We next hold that mixture fixed and choose the sampling probabilities a . Before sampling at step t , freeze AdamW’s lagged second-moment preconditioner,

$$P_t = \text{diag}((\sqrt{v_{t-1}} + \epsilon)^{-1}), \quad U_i = P_t G_i, \quad (19)$$

and define the task signal

$$s_i = \sqrt{\mathbb{E} \|U_i\|_2^2}. \quad (20)$$

This score is a lagged-preconditioned gradient norm, used as a proxy for the task-dependent AdamW step scale. It is not the exact AdamW update. The first-order loss decrease also depends on gradient direction, so we treat loss reduction as a separate empirical outcome.

For one sampled task and fixed P_t , minimizing the second moment of the importance-corrected estimator gives

$$a_i^* = \frac{w_i s_i}{\sum_j w_j s_j}. \quad (21)$$

Because Equation 4 fixes the conditional mean, this also minimizes $\mathbb{E} \|P_t \hat{g} - P_t g_w\|_2^2$. The importance-sampling minimization is classical; our modeling choice is to apply it at the fixed-token task level using the lagged AdamW proxy. The short proof is in Appendix A.5.

Gradient-Preserving Adaptive Sampling (GPAS) estimates each score online:

$$\hat{s}_i^2 \leftarrow \gamma \hat{s}_i^2 + (1 - \gamma) \|P_t G_i\|_2^2 \quad (22)$$

after sampling task i . A probability floor limits stale statistics and large importance ratios. The exact lower-bounded solution is

$$a_i = \max \left\{ a_{\min}, c w_i \sqrt{\hat{s}_i^2 + \epsilon} \right\}, \quad \sum_i a_i = 1, \quad (23)$$

where c enforces normalization. The sampled gradient is multiplied by w_I/a_I before the AdamW moment update. The name refers specifically to preserving the conditional mean of the target gradient estimator. GPAS does not preserve the expected AdamW parameter update or the optimizer trajectory. It stores one score and one timestamp per task and requires no extra model evaluation.

Raw GPAS sets $P_t = I$. Adam GPAS uses Equation 19. The latter relies on the quality of the lagged approximation, which we test in Section 5.3. Importance correction preserves the conditional expected gradient of the chosen MOPD objective, but AdamW’s second-moment state still depends on the proposal through squared importance-corrected gradients. Appendix A.4 gives the state recursion, and our experiments separately measure this order effect.

5 EXPERIMENTS

The experiments directly test the three components of our theory in Section 4. The first experiment tests the local gap diagnostic and the predicted response to task weights. The second tests whether AdamW state changes the scale of task updates. The third evaluates GPAS under a fixed teacher mixture and matched training budgets.

5.1 SETUP

We use Qwen3-1.7B as the shared initialization. Four same-origin teachers are trained from this checkpoint with single-task GRPO on mathematics, code, instruction following, and science question answering. The MOPD student starts from the same shared checkpoint. All student methods use the same frozen teachers, prompt pools, sampled-token reverse-KL loss, AdamW configuration, and learning-rate schedule.

One optimizer update uses about 65,536 valid response tokens. For task-homogeneous methods, every update contains one task. The task loss is the mean over its valid tokens. Each MOPD run uses 600 optimizer updates, including a 16-update initialization period with four observations from every task. Teacher loss is measured every 25 updates on a fixed held-out prompt set. Formal capability evaluations are run at updates 0, 150, 300, 450, and 600.

The training runs use seed 42. Prompt-level paired bootstrap intervals measure uncertainty in teacher-loss and benchmark evaluation. Every table and figure labels the result as one training seed. Appendix tables report prompt counts, valid response tokens, teacher-scored tokens, GPU hours, truncation, and verifier errors.

5.2 CAN THE LOCAL MODEL PREDICT TEACHER TRADE-OFFS?

At five checkpoints, we cache student rollouts and teacher log probabilities for the sampled response tokens of every task. The fit cache and held-out cache contain separate tokens. We solve the joint and four task-specific problems in Equation 11 with Jacobian–vector products and a matrix-free, damped trust-region conjugate-gradient solver. The radius equals the task-dependent parameter displacement of a matched 16-update branch after subtracting a zero-gradient replay from the same model and AdamW state. This removes movement due only to existing momentum and weight decay. We report $\text{SoloGap}(R)$, $\text{ConflictGap}(R)$, and $\text{OptGap}(R)$ on the held-out cache, together with their sum and the trust-region constraint status, damping, iteration count, and solve residual. The algebraic decomposition is exact on the fit objective. Held-out values are cross-fitted estimates and may show small signed deviations from the exact nonnegativity relations because of finite-cache error.

Before interpreting the gaps, we test the local approximation itself. Along each realized update direction, we evaluate the held-out sampled-token reverse-KL estimator and the quadratic surrogate at $\theta_0 + \alpha \Delta_{\text{train}}$ for $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$. We report relative approximation error and the error as a function of α alongside the gap measurements. This check tests the update scale over which Equation 7 is a useful empirical model.

For the weight response, we use Equation 17 when the joint solution is interior and Equation 39 when it is on the boundary. Because both equations predict the response of the best local fit whereas a 16-update AdamW branch may not reach that fit, we also report response accuracy as a function of $\text{OptGap}(R)$.

We test the weight-response prediction from checkpoints of uniform MOPD. For each checkpoint and each task, we add +0.5 or −0.5 to that task’s weight logit. Every branch replays the same 16 cached task batches in the same order from the same model and AdamW state. The design creates 40 branches, or 20 symmetric perturbation pairs: five checkpoints, four perturbed tasks, and two perturbation directions. The odd component of each pair measures the predicted first-order response; the even component measures departure from local linearity.

Equation 18 predicts the change in all four task losses. We compare prediction and observation with sign accuracy, Spearman correlation, normalized error, and a task-by-task response heatmap. The main test uses the full predicted response matrix. The relation between the conflict gap and the size of the observed trade-off is reported as a supporting diagnostic.

For every perturbed weight vector, we solve the corresponding local joint problem to obtain a branch-specific optimization gap. This distinguishes failure of the weight-response approximation from a finite branch that simply does not approach its predicted local fit.

The matched-branch experiment connects each local quantity to an observed training outcome. The cross-fitted solo residual measures taskwise local fit inside the tested radius. The sharing penalty measures the extra residual from requiring one update, and the optimization residual measures the improvement available to a better joint update in the same local model. None of these quantities is interpreted as global student capacity. The response matrix predicts which task losses move when the target mixture changes.

5.3 DOES ADAMW STATE CHANGE TASK UPDATE SCALE?

We isolate optimizer state with same-checkpoint one-step clones. At each of five uniform-MOPD checkpoints, every task provides 16 independent fixed-token batches. Each of the 320 observations starts from the same copied model, first-moment state, and second-moment state for that checkpoint.

For each observation, we record the raw gradient norm, the lagged- preconditioned norm, and the exact AdamW parameter step. We remove the common movement from existing momentum by subtracting the step produced by a zero-gradient AdamW clone:

$$\Delta_i^{\text{task}} = \Delta_i^{\text{AdamW}} - \Delta^{\text{zero}}. \quad (24)$$

The primary test compares both gradient scores with $\|\Delta_i^{\text{task}}\|_2$. We report correlation, pairwise task-ranking accuracy, calibration slope, and checkpoint-stratified results. Held-out teacher-loss reduction is a secondary outcome because loss reduction also depends on gradient direction.

We measure task order in a separate replay. From update 300, four schedules use the same 64 cached task batches: random shuffle, round robin, one contiguous task order, and the reverse contiguous order. Final per-task loss, parameter displacement, and AdamW moment states measure the effect of order under identical task counts. SGD and frozen-preconditioner replays provide controls in the appendix.

5.4 DOES GPAS IMPROVE FIXED-MIXTURE MOPD?

All allocation methods use target weights $w_i = 1/4$. We compare the following training rules under the same target response-token and teacher-scored-token budgets:

1. **Uniform**: one task per update with $a_i = 1/4$.
2. **Full-mixture accumulation**: every AdamW step assigns one quarter of its token budget to each task, accumulates the four normalized task gradients, and applies $\sum_i w_i G_i$.
3. **Gap-adaptive**: task sampling follows the sampled teacher-student gap and uses the sampling probabilities as effective task weights.
4. **Raw GPAS**: Equation 23 uses raw gradient RMS and applies w_I/a_I .
5. **Adam GPAS**: Equation 23 uses the lagged-preconditioned RMS and applies w_I/a_I .
6. **Uncorrected Adam allocation**: the proposal matches Adam GPAS and the sampled gradient uses unit scale. The paired ablation isolates the importance correction.

The full-mixture baseline measures the value of removing task-level sampling noise before the AdamW update and provides a strong reference for the one-task-per-update setting where GPAS is designed to operate.

We measure optimization efficiency with held-out weighted teacher loss versus valid response tokens, teacher-scored tokens, and wall-clock time. Summary metrics include normalized loss area, tokens required to reach 80% and 90% of the uniform run’s final improvement, and the final optimization gap.

At each main checkpoint, independent task gradients estimate the full target gradient g_w . The same gradients measure

$$\mathbb{E} \|P_t \hat{g} - P_t g_w\|_2^2 \quad (25)$$

for every proposal. The estimator-error measurement directly evaluates the sampling result in Equation 21.

We evaluate teacher-mixture drift with the four-task teacher-loss vector at a matched value of weighted teacher loss. Final benchmark vectors provide a capability-level check. The comparison between Adam GPAS and its uncorrected ablation isolates target-gradient preservation. The comparison between Raw and Adam GPAS tests whether the optimizer-scaled score improves end-to-end training after it improves the one-step update prediction in Section 5.3.

6 PRACTICAL TRAINING GUIDE AND SCOPE

The analysis suggests a short MOPD workflow.

1. Define the teacher-loss mixture before choosing the sampler. Choose task weights to encode the intended teacher-loss objective, informed by deployment priorities. These weights need not equal deployment traffic shares unless the task losses have been calibrated for that interpretation. Average each task loss over valid response tokens before applying its task weight. Use fixed-token task batches when response lengths differ strongly.

2. Start with uniform task sampling and measure local gaps. At a few checkpoints, estimate the solo, conflict, and optimization gaps on held-out cached tokens. A large solo residual means that a task-specific linearized update does not fit the cached teacher signal within the tested radius; it is not by itself evidence of insufficient model capacity. A large sharing penalty indicates that one local update fits worse than separate task-specific updates. A large optimization residual points to allocation or training budget within the same local model.

3. Use the weight response for mixture changes. Short matched branches can verify the predicted per-task loss response before committing to a long run. The short-branch check is especially useful when one teacher has strong effects on tasks outside its routed domain.

4. Use GPAS for an optimization-limited run. Raw GPAS is the direct importance-sampling rule. Adam GPAS is useful when the lagged-preconditioned norm predicts the realized task-dependent AdamW step better than the raw norm. Set the probability floor from a maximum acceptable importance ratio and a maximum acceptable gap between task observations. Random task interleaving keeps long task blocks from dominating AdamW’s recent moment history.

Scope. The local model freezes student prefixes and linearizes output changes around one checkpoint. Long training changes the prefix distribution, Jacobian, teacher gap, and optimizer state. Repeated measurements describe this moving training process more accurately than one global extrapolation. The Euclidean update radius is a coordinate-dependent, budget-matching device for one fixed model parameterization. It should not be compared across model architectures or interpreted as an intrinsic distance between policies.

The experiments use the sampled-token reverse-KL estimator and a fixed task taxonomy. Different targets, such as deployment traffic, example frequency, worst-task quality, or unequal inference cost, require matching task weights and sampling units. The main training comparison uses one model family and one training seed; prompt-level intervals quantify evaluation uncertainty for these runs.

GPAS optimizes the second moment of a one-step estimator in a chosen local metric and preserves the conditional mean of the target gradient. It does not preserve the expected AdamW update or training trajectory. Long-horizon loss and wall-clock time also depend on rollout cost, gradient direction, and AdamW state. The end-to-end experiments measure these outcomes separately.

7 CONCLUSION

Multi-teacher OPD combines several teacher requests in one shared student. A remaining teacher loss can reflect per-task reachability, additional conflict from sharing one update, or unfinished opti-

mization. Our local diagnostic separates these sources and predicts how a small task-weight change moves teacher loss between tasks.

The gap diagnosis also clarifies the roles of weights and sampling. Task weights define the intended teacher-loss mixture. Sampling probabilities allocate training updates. GPAS uses optimizer-scaled gradient signals to allocate those updates and applies an importance correction to preserve the conditional target gradient. The experiments test each link in this chain: local loss response, optimizer-dependent update scale, estimator error, training efficiency, and teacher-mixture drift.

AI USE STATEMENT

We used generative AI tools for language editing and manuscript organization. The authors reviewed every AI-assisted revision and take responsibility for the final text, claims, proofs, code, and experimental results.

ETHICS STATEMENT

The study evaluates optimization methods for existing language models with established reasoning and instruction-following datasets and automated verifiers. Model and dataset biases can affect the reported capability scores. We record data provenance, invalid outputs, and verifier failures to support careful interpretation of the results.

REPRODUCIBILITY STATEMENT

Section 5 specifies the model family, task setup, update budget, checkpoints, and evaluation protocol. The appendix gives the complete derivations. The supplementary experiment record includes frozen data manifests, model and tokenizer revisions, random seeds, optimizer settings, task-sampling logs, checkpoint states, and per-prompt evaluation outputs.

REFERENCES

- Guillaume Alain, Alex Lamb, Chinnadhurai Sankar, Aaron Courville, and Yoshua Bengio. Variance reduction in sgd by distributed importance sampling, 2015. URL <https://arxiv.org/abs/1511.06481>.
- Zhao Chen, Vijay Badrinarayanan, Chen-Yu Lee, and Andrew Rabinovich. Gradnorm: Gradient normalization for adaptive loss balancing in deep multitask networks. In *Proceedings of the 35th International Conference on Machine Learning*, pp. 794–803, 2018. URL <https://proceedings.mlr.press/v80/chen18a.html>.
- Nicholas E. Corrado, Wenyuan Huang, and Josiah P. Hanna. Distributionally robust multi-task reinforcement learning via adaptive task sampling, 2026. URL <https://arxiv.org/abs/2605.14350>.
- Huan-ang Gao, Haohan Chi, Yong Yan, Shiyuan Feng, Hanlin Wu, Zheng Jiang, Bingxiang He, Wei-Ying Ma, Ya-Qin Zhang, and Hao Zhou. Open-mopd: Diagnosing and fixing capability imbalance in multi-teacher on-policy distillation, 2026. URL <https://arxiv.org/abs/2608.19098>.
- Angelos Katharopoulos and François Fleuret. Not all samples are created equal: Deep learning with importance sampling. In *Proceedings of the 35th International Conference on Machine Learning*, pp. 2525–2534, 2018. URL <https://proceedings.mlr.press/v80/katharopoulos18a.html>.
- Yaxuan Li, Yuxin Zuo, Bingxiang He, Jinqian Zhang, Chaojun Xiao, Cheng Qian, Tianyu Yu, Huan-ang Gao, Wenkai Yang, Zhiyuan Liu, and Ning Ding. Rethinking on-policy distillation of large language models: Phenomenology, mechanism, and recipe, 2026a. URL <https://arxiv.org/abs/2604.13016>.

Zhaoyi Li, Deyang Kong, Yuan Wei, Evan Yang, Ranran Shen, Mahardika Krisna Ihsani, Ming Yang, Wei Zhang, Chuan Hao, Jian Yang, Ran Tao, Bryan Dai, Shikun Zhang, Wei Ye, Ying Wei, and Defu Lian. Every coin has two sides: On the dual nature of generalization in on-policy distillation of large language models, 2026b. URL <https://arxiv.org/abs/2608.16647>.

Wenhan Ma, Jianyu Wei, Liang Zhao, Hailin Zhang, Bangjun Xiao, Lei Li, Qibin Yang, Bofei Gao, Yudong Wang, Rang Li, Jinhao Dong, Zhifang Sui, and Fuli Luo. Mopd: Multi-teacher on-policy distillation for capability integration in llm post-training, 2026a. URL <https://arxiv.org/abs/2606.30406>.

Yingzi Ma, Zichen Zhu, Ming Jiang, and Chaowei Xiao. One student, many teachers: Multi-task on-policy distillation via soft-prompt privileged context, 2026b. URL <https://arxiv.org/abs/2607.18293>.

Yuan Xie, Jiaqi Song, Xianliang Wang, Ming Lei, Jie Gao, and Jie Wu. Language-specialized multi-teacher on-policy distillation for multilingual llm-based asr, 2026. URL <https://arxiv.org/abs/2608.03610>.

Kejian Zhu, Zhuoran Jin, Shangqing Tu, Hongbang Yuan, Yushi Bai, Kang Liu, Juanzi Li, and Jun Zhao. Sft conflicts, rl coexists: A theoretical and empirical analysis of multi-task learning for llms, 2026. URL <https://arxiv.org/abs/2608.03573>.

A PROOFS AND ADDITIONAL DERIVATIONS FOR OUR THEORY

A.1 CONSTRUCTION OF THE SAMPLED-TOKEN LOCAL SURROGATE

Fix one cached valid response token $z = (c, y)$, where the rollout token was sampled from the checkpoint student. Define

$$d_i(z) = \log \tau_i(y | c) - \log \pi_{\theta_0}(y | c), \quad j_i(z) = \nabla_{\theta} \log \pi_{\theta_0}(y | c). \quad (26)$$

On this fixed cache, Taylor expansion in parameter space gives

$$\log \pi_{\theta_0 + \Delta}(y | c) = \log \pi_{\theta_0}(y | c) + j_i(z)^{\top} \Delta + O(\|\Delta\|^2). \quad (27)$$

Thus $d_i(z) - j_i(z)^{\top} \Delta$ is the first-order teacher-student log-probability residual for that sampled token. Squaring and averaging these residuals gives Equation 7. For a cache $\mathcal{C}_i = \{z_{i,n}\}_{n=1}^{N_i}$, the objective used by the solver is

$$\hat{L}_w(\Delta) = \frac{1}{2} \sum_{i=1}^M \frac{w_i}{N_i} \sum_{n=1}^{N_i} (d_i(z_{i,n}) - j_i(z_{i,n})^{\top} \Delta)^2. \quad (28)$$

The implementation evaluates the linear maps $\Delta \mapsto \{j_i(z_{i,n})^{\top} \Delta\}_n$ and their adjoints with automatic differentiation. It never stores a token-by-parameter Jacobian or a parameter-by-parameter curvature matrix.

This construction is deliberately aligned with sampled-token OPD (Ma et al., 2026a; Li et al., 2026a). At one prefix, the reverse KL is $\mathbb{E}_{y \sim \pi_{\theta}} [\log \pi_{\theta}(y | c) - \log \tau_i(y | c)]$, so its usual Monte Carlo estimator uses exactly the log-ratio appearing above. A standard second-order expansion about equality of the student and teacher distributions yields a weighted squared log-ratio, which motivates the local least-squares geometry. Equation 28 does not require uniform closeness over the full vocabulary, however, and we do not identify it with the exact KL away from that local regime. The interpolation check in Section 5.2 tests whether the surrogate is accurate over the update scale used by the diagnostic.

A.2 PROOF OF THE LOCAL-FIT DECOMPOSITION AND WEIGHT RESPONSE

Write $\mathbb{E}_i[\cdot]$ for expectation over $z \sim \nu_i^{\theta_0}$. Define the positive-semidefinite operator

$$H_w = \sum_{i=1}^M w_i \mathbb{E}_i [j_i(z) j_i(z)^{\top}], \quad b_w = \sum_{i=1}^M w_i \mathbb{E}_i [j_i(z) d_i(z)]. \quad (29)$$

Differentiating equation 7 gives

$$H_w \Delta \theta_w^* = b_w. \quad (30)$$

This is operator notation: neither H_w nor an inverse is formed. Other minimizers differ by $v \in \text{Null}(H_w)$, and

$$v^\top H_w v = \sum_i w_i \mathbb{E}_i [(j_i(z)^\top v)^2] = 0 \quad (31)$$

implies $j_i(z)^\top v = 0$ almost surely, so all minimizers give the same fitted first-order change in sampled-token log probabilities.

Let $e_{i,w}^*(z) = d_i(z) - j_i(z)^\top \Delta \theta_w^*$. The normal equation gives

$$\sum_i w_i \mathbb{E}_i [j_i(z) e_{i,w}^*(z)] = 0. \quad (32)$$

Expanding around a minimizer gives

$$L_w(\Delta \theta) = \frac{1}{2} \sum_i w_i \mathbb{E}_i [(e_{i,w}^*(z) - j_i(z)^\top (\Delta \theta - \Delta \theta_w^*))^2] \quad (33)$$

$$= L_w^* + \frac{1}{2} \sum_i w_i \mathbb{E}_i [(j_i(z)^\top (\Delta \theta - \Delta \theta_w^*))^2], \quad (34)$$

because equation 32 cancels the cross term, proving Proposition 1. Moreover, $L_w^* = 0$ exactly when one $\Delta \theta$ satisfies $j_i(z)^\top \Delta \theta = d_i(z)$ for every task almost surely.

For the constrained diagnostic, every feasible shared update obeys

$$\sum_i w_i \ell_i(\Delta) \geq \sum_i w_i \min_{\|\Delta_i\|_2 \leq R} \ell_i(\Delta_i). \quad (35)$$

Taking the minimum over the shared update proves $L_w^*(R) \geq L_{w,\text{solo}}^*(R)$ and hence the nonnegativity of $\text{ConflictGap}(R)$. Feasibility of Δ_{train} similarly gives $L_w(\Delta_{\text{train}}) \geq L_w^*(R)$, proving nonnegativity of $\text{OptGap}(R)$. The three-part identity in Equation 15 follows by cancellation.

For $w = \text{softmax}(\eta)$, first suppose the interior solution is unique on the solver subspace. Differentiating equation 30 gives

$$\frac{\partial \Delta \theta_w^*}{\partial \eta_k} = H_w^{-1} w_k \mathbb{E}_k [j_k(z) e_{k,w}^*(z)] = H_w^{-1} w_k r_{k,w}^*. \quad (36)$$

The residual gradient supplies the task-specific response, and H_w^{-1} maps it through shared curvature. In the matrix-free implementation, we solve $(H_w + \xi I)u_k = w_k(r_{k,w}^* - \xi \bar{\Delta}_w)$ by conjugate gradients and report the damping ξ and solve residual. The additional term follows from $\sum_i w_i r_{i,w} = \xi \bar{\Delta}_w$ at the damped optimum. This is the sensitivity of the same damped local problem and avoids forming an inverse or selecting a formal pseudoinverse of a massively overparameterized model.

For completeness, suppose the radius is active. Write $\bar{\Delta}_w = \arg \min_{\|\Delta\|_2 \leq R} L_w(\Delta)$ and let $\lambda_w > 0$ be the multiplier for $\frac{1}{2}(\|\Delta\|^2 - R^2) \leq 0$. The KKT conditions are

$$(H_w + \lambda_w I) \bar{\Delta}_w = b_w, \quad \|\bar{\Delta}_w\|^2 = R^2. \quad (37)$$

Using $\partial w_j / \partial \eta_k = w_j(\mathbf{1}\{j = k\} - w_k)$ and differentiating both conditions gives

$$(H_w + \lambda_w I)u_k + \bar{\Delta}_w \dot{\lambda}_k = w_k(r_k - \lambda_w \bar{\Delta}_w), \quad (38)$$

$$\bar{\Delta}_w^\top u_k = 0. \quad (39)$$

which is Equation 39. Thus the weight-response prediction remains aligned with the trust-region problem used in the experiment.

A.3 OPTIMIZER-AWARE SHORT-HORIZON APPROXIMATION

The objective determines the projection. The optimizer controls the finite-step path. For $\hat{g} = (w_I/a_I)G_I$,

$$\mathbb{E}[\hat{g}] = \sum_i w_i \mathbb{E}[G_i]. \quad (40)$$

For a one-step sampling decision, we freeze AdamW’s preconditioner at its pre-sampling value. Up to task-independent momentum and weight decay, the task-dependent update is

$$\Delta\theta_{t,I} \approx -\alpha_t(1 - \beta_1)P_t\hat{g}, \quad P_t = \text{diag}((\sqrt{v_{t-1}} + \epsilon)^{-1}). \quad (41)$$

Bias correction is shared by all tasks and leaves their relative sampling probabilities unchanged. This linearized update motivates the preconditioned scale. Exact AdamW remains nonlinear:

$$\mathbb{E}[\text{AdamW}(\hat{g})] \neq \text{AdamW}(\mathbb{E}[\hat{g}]) \quad (42)$$

because the current gradient also changes the second-moment state. We test the lagged approximation in Section 5.3.

A.4 TASK ORDER AND EXACT ADAMW STATE DYNAMICS

Let $\mathcal{F}_{t-1}$ denote the student, optimizer state, cached data, and task history before step t . Choose a_t from this history, draw $I_t \sim a_t$, and define

$$H_t = \rho_{t,I_t} G_{t,I_t}, \quad \rho_{t,i} = \frac{w_i}{a_{t,i}}, \quad (43)$$

where $G_{t,i}$ is the current mean-token gradient for task i . Then

$$\mathbb{E}[H_t \mid \mathcal{F}_{t-1}] = \sum_i w_i g_{t,i} = g_w(\theta_t). \quad (44)$$

AdamW updates its states and parameters as

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) H_t, \quad (45)$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) H_t^{\odot 2}, \quad (46)$$

$$\theta_{t+1} = (1 - \alpha_t \lambda) \theta_t - \alpha_t \frac{\tilde{m}_t}{\sqrt{\tilde{v}_t} + \epsilon}, \quad (47)$$

where $\tilde{m}_t, \tilde{v}_t$ are bias corrected and the last line is coordinatewise. With $q_{t,i} = \mathbb{E}[G_{t,i}^{\odot 2} \mid \mathcal{F}_{t-1}]$,

$$\mathbb{E}[m_t \mid \mathcal{F}_{t-1}] = \beta_1 m_{t-1} + (1 - \beta_1) g_w(\theta_t), \quad (48)$$

$$\mathbb{E}[v_t \mid \mathcal{F}_{t-1}] = \beta_2 v_{t-1} + (1 - \beta_2) \sum_i \frac{w_i^2}{a_{t,i}} q_{t,i}. \quad (49)$$

Importance weighting removes a_t from the expected first-moment increment. The expected second-moment increment retains a_t . Even $a_t = w$ differs from a deterministic full-mixture AdamW step, whose squared gradient contains cross terms.

The realized state also depends on task order. For a fixed sequence of importance-weighted gradients, unrolling the recursions gives

$$m_T = \beta_1^T m_0 + (1 - \beta_1) \sum_{t=1}^T \beta_1^{T-t} H_t, \quad (50)$$

$$v_T = \beta_2^T v_0 + (1 - \beta_2) \sum_{t=1}^T \beta_2^{T-t} H_t^{\odot 2}. \quad (51)$$

These recency-weighted sums generally change under permutation, even with frozen gradients; in MOPD, earlier updates also change later parameters, rollouts, and gradients. Random or predictable adaptive sampling retains equation 44, while round-robin and blocked schedules match aggregate counts and lack per-step conditional unbiasedness. In all cases the objective-level projection is unchanged, while the finite-step path is order dependent. Our implementation applies w_I/a_I before both moment updates and uses random task interleaving by default.

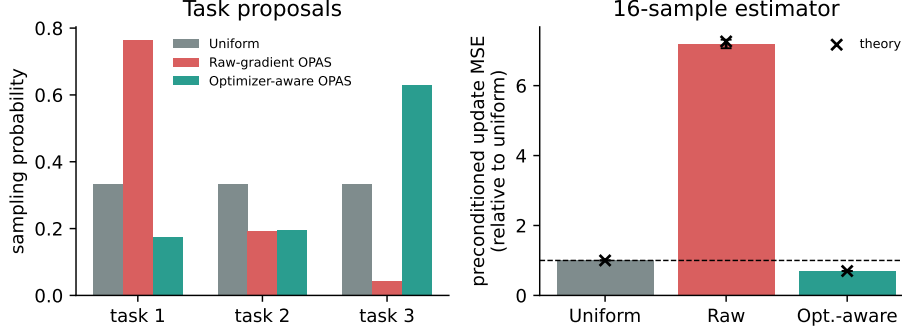


Figure 1: Controlled check of optimizer-aware task sampling. Crosses show the theoretical mean-squared error (MSE); bars show 20,000 Monte Carlo trials. Sampling by raw gradient norm over-allocates the task whose large coordinates are suppressed by the optimizer, producing $7.19\times$ the uniform MSE. Optimizer-aware GPAS matches the predicted proposal and reduces MSE to $0.698\times$ uniform.

A.5 OPTIMAL GRADIENT-PRESERVING SAMPLING

Conditional on the lagged preconditioner P_t , let $s_i^2 = \mathbb{E}\|P_t G_i\|_2^2$. The second moment of the preconditioned importance-weighted gradient is

$$\mathbb{E}\|P_t \hat{g}\|_2^2 = \sum_{i=1}^M \frac{w_i^2 s_i^2}{a_i}. \quad (52)$$

For any distribution a , Cauchy–Schwarz gives

$$\left(\sum_i w_i s_i \right)^2 = \left(\sum_i \sqrt{a_i} \frac{w_i s_i}{\sqrt{a_i}} \right)^2 \quad (53)$$

$$\leq \left(\sum_i a_i \right) \left(\sum_i \frac{w_i^2 s_i^2}{a_i} \right). \quad (54)$$

Equality holds exactly when a_i is proportional to $w_i s_i$. Normalization gives

$$a_i^* = \frac{w_i s_i}{\sum_j w_j s_j}. \quad (55)$$

The equality condition proves equation 21; the minimum is $(\sum_i w_i s_i)^2$. With $P_t = I$, the optimal proposal reduces to raw-gradient importance sampling.

With a lower bound $a_i \geq a_{\min}$, the Karush–Kuhn–Tucker conditions give

$$a_i = \max\{a_{\min}, c w_i s_i\}, \quad \sum_i a_i = 1. \quad (56)$$

The scalar c is unique when $M a_{\min} < 1$ and at least one score is positive. Equation 23 replaces s_i with its moving estimate.

A.6 CONTROLLED OPTIMIZER-AWARE SAMPLING CHECK

Before running full MOPD, we verify the optimizer-aware sampling claim in a controlled local-gradient experiment. Three equally weighted tasks have noisy gradients concentrated in different parameter blocks, and a fixed diagonal preconditioner rescales those blocks differently. Every method estimates the target gradient from batches of 16 sampled tasks. The controlled experiment tests the conditional second-moment result in equation 52. The experiment is a local gradient simulation, followed by a separate LLM training evaluation in the main paper.

The agreement between theory and simulation confirms the conditional variance calculation and demonstrates why raw gradient norm can be the wrong allocation signal. The full MOPD experiments in Section 5 remain necessary to determine whether optimizer-aware sampling improves end-to-end training.